

FT85RS With Enclosure Manual and Features V1.0

Specifications :

- Firmware: V1.5 and above (firmware update via Flipsky ESC Tool supported)
- Voltage: 14V-85V (safe for 4S-20S)
- Continuous current: 50V/550A; 75V/450A; 85V/400A(Values depend on the mounting, ambient temperature and air/water circulation around the device.)
- Peak Current: 1000A
- BEC: 5V@1A
- Power/Motor Wires: Integrated M8 terminals
- Programmable Tool: Flipsky ESC TOOL
- PWM Frequency Control: YES
- Multiple Speed Gear Limit Control: YES (UART / ADC)
- Regenerative Capacity: Yes
- Communication Ports: USB, CAN, UART, PPM, ADC/PAS
- Input Signals Support: PPM, ADC, PAS, UART(compatible with FLIPSKY remote such as VX1, VX1Pro, VX2, VX3, VX3Pro, VX4, VX4Pro)
- Startup Modes Support: IPDS (sensorless), HFPI (sensorless), HALL (sensored), ENCODER (ABI Encoder, AS5047 Absolute Type Encoder SPI Port), OPEN(Sensorless Open-Loop Startup)
- Traction Control: YES
- Multiple Speed Gear levels: Low, Medium, High, Reverse
- Switch Support: Power on/off button switch or key switch
- Size:L218*W134*H54.8(with aluminum enclosure)

Features:

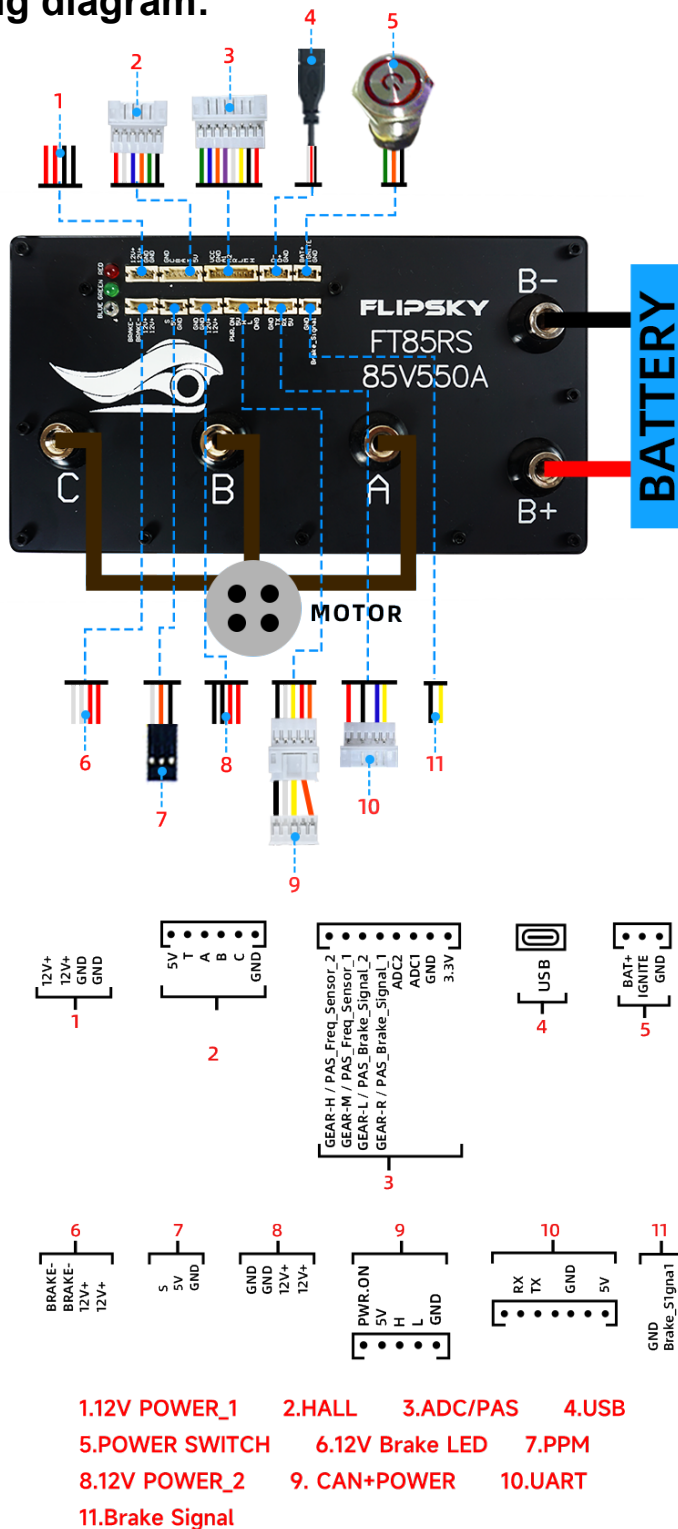
- Adopting ARM Cortex-M4 structure with the max 240MHz , improves PWM switching frequency and the motor RPM.
- Programmable via Flipsky ESC Tool to be compatible with multi-types of motor, different battery voltages and multiple control methods.
- Programmable via Flipsky remote controller VX4 and VX4Pro(remote firmware version: VX4_V1.4.1 and above; VX4Pro_V1.1 and above)
- Supporting motor FOC with low working noise and smooth starting.
- Supporting Real-time data and fault alerts.
- Supporting adaptive variable PWM frequency control.
- Supporting sensor mode(HALL, ENCODER) and sensorless mode(IPDS, HFPI, OPEN).
- Supporting control types: Current, Duty Cycle, Speed Control, Position Control.

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- Supporting multiple types of motor temperature sensors such as NTC-10k-25℃, NTC-10K-100℃, PT1000, PTC-1K-100℃, KTY, etc.
- Protection functions: low and high voltage protection, motor over-temperature protection, MOSFET over-temperature and over current protection, battery over discharge protection.
- Integrating power of 12V(3pcs) to provide power for external devices(headlight, brake light, horn, etc.) with two groups normally open and one controlled by brake signal. (Note: maximum output current:4A)
- Supporting brake signal to cut off ESC output.

Wiring diagram:



To download the Flipsky ESC Tool and User Manual, please scan the QR code:



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FT85RS 外壳版规格书及产品特点（版本 V1.0）

规格参数:

- 固件：V1.5 及以上，支持通过 Flipsky ESC Tool 升级
- 电压：14V-85V (安全范围：4S-20S)
- 持续电流：50V/550A 75V/450A 85V/400A（持续时间取决于外部散热情况）
- 峰值电流：1000A
- BEC:5V@1A
- 电力线缆：集成 M8 端子
- 电机线缆：集成 M8 端子
- 调参工具：Flipsky ESC TOOL
- PWM 变频控制：YES
- 多挡位限速控制: YES(UART、ADC)
- 再生制动：YES
- 通讯端口：USB, CAN, UART, PPM, ADC/PAS
- 控制信号类型：PPM, ADC, PAS, UART（目前兼容 VX1、VX1Pro、VX2、VX3、VX3Pro、VX4、VX4Pro 等遥控器）
- 启动模式：IPDS(无感轻载)，HFPI(无感重载)，HALL, ENCODER（ABI编码器，AS5047绝对式编码器 SPI接口），OPEN(无感开环启动)
- 牵引力控制：YES
- 速度档位：低速档，中速档，高速档，倒车档
- 集成了自锁开关或电门锁开关
- 尺寸：长 218mm、宽 134 mm、高 54.8mm（带铝外壳）

产品特点:

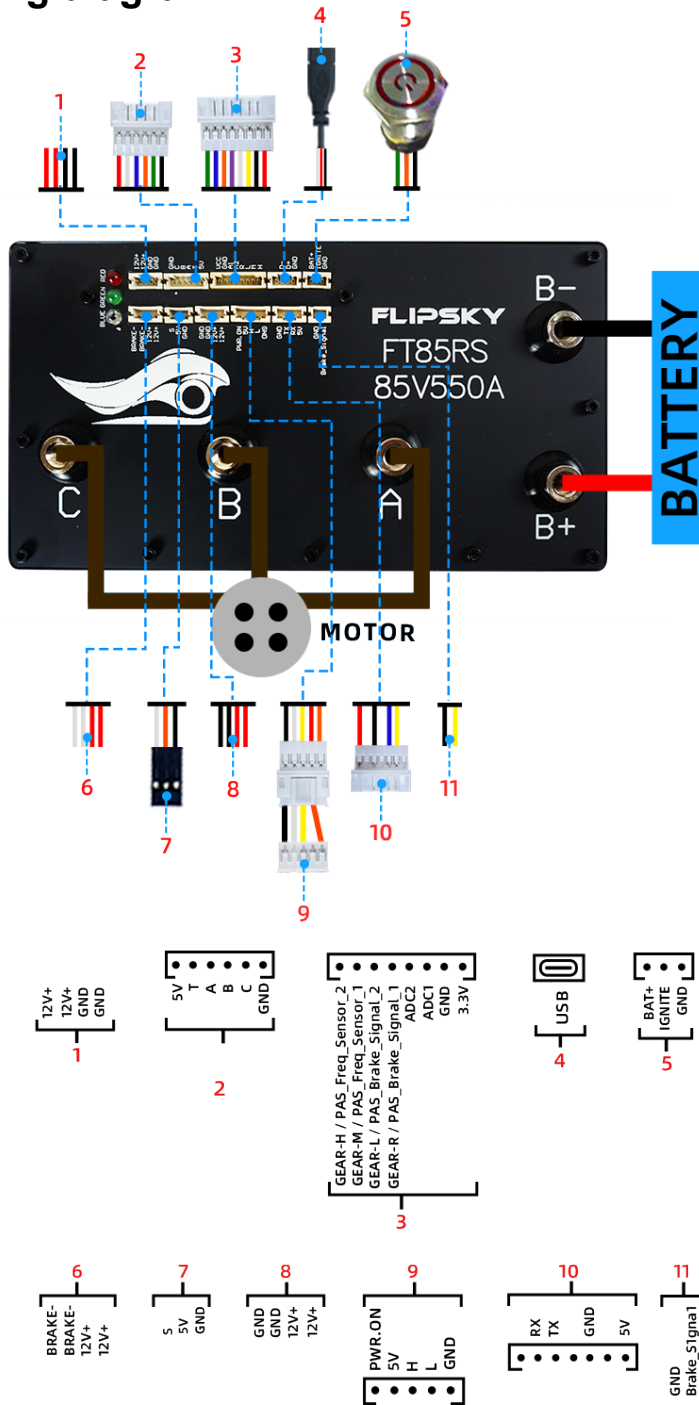
- 采用 ARM Cortex-M4 架构，高达 240MHz 主频，可提高 PWM 控制频率和电机转速上限；
- 可通过 Flipsky ESC Tool 编程，兼容多种电机、不同电池电压和多种控制方式；
- 支持 Flipsky VX4 与 VX4Pro 遥控器调参（固件版本：VX4_V1.4.1 及以上版本;VX4Pro_V1.1 以上版本）；
- 支持 FOC，工作噪音低，启动平稳；
- 支持实时数据和故障报警支持自适应 PWM 变频控制；
- 支持传感器模式（HALL、ENCODER）和无传感器模式（IPDS、HFPI、OPEN）；
- 支持控制类型：电流（Current）、占空比（Duty Cycle）、速度控制（Speed Control）、位置控制（Position Control）；
- 保护功能：低压保护；高压保护；电机过温保护；MOS 管过流保护；电池过放保护；MOS 管过温保护；

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- 支持多种类型电机温度传感器，包括 NTC-10k-25°C、NTC-10K-100°C、PT1000、PTC-1K-100°C、KTY 系列等多种类型温度传感器。
- 集成 3 组 12V 功率电源，用于外接设备供电（如前照灯、刹车灯、喇叭装置等等），其中一组是通过刹车信号控制开启，其它两组则是常开状态；注：12V 电源最大输出电流 4A；
- 支持刹车信号（Brake Signal）切断电调输出。

Wiring diagram:



- 1.12V POWER_1 2.HALL 3.ADC/PAS 4.USB
5.POWER SWITCH 6.12V Brake LED 7.PPM
8.12V POWER_2 9. CAN+POWER 10.UART
11.Brake Signal

Flipsky ESC TOOL
软件+中文用户手册
下载二维码：



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